

orbit_demo review notes: issues, trade-offs, questions, discrepancy register

Package section 09_review_notes of orbit_demo_design_review, written 2026-10-06. Links are relative to this file. Paths in backticks without a link (reports/..., build/..., site/..., viz/..., docs/research/..., ORFS image paths) are repository, build or tool-image paths in /home/user/chip that are not copied into the package. "repo" = /home/user/chip.

Status labels (one per result): **VERIFIED** = a tool log or report shows it. **REPRODUCED 2026-10-06** = re-run or re-checked for this package. **TARGET** = design goal, not measured. **ASSUMPTION. CLAIM (unverified)** = stated in a document, no log found. **MISSING. N/A** (reason given).

Severity: **High** = blocks sign-off or makes a published statement wrong. **Medium** = limits what the evidence shows; owner action needed before sign-off. **Low** = wording, labelling or traceability.

0. Revision covered

The package revision table, verbatim:

Item	Revision / identifier	Note
Design	orbit_demo (ORBIT-AI 4-lane INT8 digital demonstrator)	top module orbit_demo, LANES=4
RTL	rtl/orbit_demo.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_keep_reg.v at git commit 26566f0 (2026-09-29 18:54 UTC); unchanged through the package commit	sha256 a3fffb22... / efef3d33... / bba77a5a... / ea05bdc1... (full hashes in MANIFEST.sha256)
Specification	docs/SPEC.md at 26566f0 (never revised)	sha256 aef5d9fb...
Concept brief	docs/orbit-ai-design-brief.pdf, "ORBIT-AI v0.1, 29 September 2026" (never revised; describes an earlier project state, see discrepancy register)	sha256 9d5f33d3...
Reviewed layout	ORFS sky130hd, variant sep (copy-separation fences), run 2026-09-29 21:56-22:12 UTC, outputs build/pd_sep/results/sky130hd/orbit_demo/sep/	6_final.gds sha256 cd19afa9...; 6_final.def f5c544f3...; 6_final.v 68093c41...; 6_final.spef 014e655a...
Layout configuration	pd/sky130hd_sep/{config.mk, constraint.sdc, regions.tcl} at c10c59b	The run used the ed553f8 versions; with comments stripped they are identical to c10c59b (comment-only changes)
Flow / tools	ORFS docker image openroad/orfs@sha256:2e5bf6fe865e... (created 2026-09-29 02:00 UTC); OpenROAD prints version "unknown"; KLayout 0.30.12 (DRC/LVS); Yosys 0.69+154 (local sims/schematics)	No PDK commit is recorded in the logs
Process / library	SkyWater SKY130, sky130_fd_sc_hd, liberty sky130_fd_sc_hd_tt_025C_1v80 (the single corner ORFS optimised at)	—
Package assembled	2026-10-06 from repo commit 0495cfa (branch claude/hopeful-rubin-0yf8io)	Re-runs made for this package are dated 2026-10-06

Other implementations in the repo (sky130hd baseline at 7.0 ns, variant base; IHP SG13G2) are reference runs only and are NOT the reviewed layout.

Corrections found while checking this table: the "ed553f8 versions" note is wrong for constraint.sdc (D-52); MANIFEST.sha256 is not in the package (D-53; full hashes are in [04_digital_design/README.md](#) section 1); the checkout has moved past 0495cfa (D-54).

1. Known issues and limitations, by severity

ID	Sev.	Issue and evidence	Source (package; repo/build if different)	Conditions	Status
K-01	High	Setup fails at ss_100C_1v60 on the reviewed layout. Setup WNS -6.131 ns, TNS -2463.532 ns, 516 violating endpoints: all 512 bits of g_lane[0..3].u_lane.u_{acc_a,acc_b,res_a,res_b}/q[*], plus out_valid_q, fault_q and outputs in_ready (-4.142 ns) and out_valid (-3.838 ns). Worst path in_a[12] → g_lane[1].u_lane.u_acc_a/q[31]\$_SDFFE_PN0P_.reg2reg also fails (-4.420 ns at g_lane[0].u_lane.u_res_b/q[30]), so the failure is not only the I/O budget. Min period 13.33 ns (75.01 MHz) with I/O delay held at 1.44 ns. At SS also 37 max-slew and 12 max-capacitance violating pins. Hold is met at all corners (smallest: FF 0.281 ns). TT: WNS +0.031 ns; FF: +2.276 ns. The only committed corner report is for the baseline (-6.016 ns at 7.0 ns).	sta_corners_sep.txt , sta_corners_sep_supplement.txt , sta_ss_100C_1v60.log , sta_extra_ss_100C_1v60.log (endpoint list, 516 VIOLATED); baseline sta_corners_baseline_7p0ns.txt	OpenSTA in OpenROAD "unknown", image sha256:2e5bf6fe865e; pd/sta_corner.s.tcl unmodified; build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.{odb,sdc,spef}; 7.2 ns, I/O delay 1.44 ns, propagated clock; nominal OpenRCX SPEF at every corner; no derate/OCV/uncertainty; design optimised by ORFS at TT only	REPRODUCED 2026-10-06
K-02	High	Single points of failure. One instance, feeding both copies, of: the 8x8 multiplier and prod32 per lane (orbit_mac_lane.v:24-26); the A/B comparators (orbit_mac_lane.v:59) and the mismatch OR-tree mismatch = \ lane_mismatch (orbit_demo.v:55); the lane control in_fire → mac_en, clear_fault → clr, in_first, in_last, acc_en/res_en and the clr d-mux selects (orbit_demo.v:81-84; orbit_mac_lane.v:36-57); the thermal voter (orbit_thermal_tmr.v:40) and the one nxt cone that writes all three copies (:44-68); the unprotected u_thermal.phase (:72-78), fault_q and out_valid_q (orbit_demo.v:57-58, 107-122); the clock (one CTS tree, 521 sinks); rst_n (synchronous, no synchronizer); the sensor and handshake inputs. In the reviewed netlist, shared drivers also cross copies: clknet_leaf_22_clk clocks u_thermal.u_copy0/q[1] and u_thermal.u_copy1/q[1]; place1872 (buf_4, net net1870) drives reset gates of u_thermal.u_copy1, u_thermal.u_copy2, g_lane[2].u_lane.u_acc_a, g_lane[2].u_lane.u_res_a and g_lane[3].u_lane.u_acc_a. SPEC section 8 lists only part of this (D-01).	RTL in source/rtl/SPEC.md 151-162; 4_1_cts.log ; netlist 6_final.v.gz ; summary.md 194-195	RTL at 26566f0; netlist = sep_6_final.v (sha256 68093c41...), parsed 2026-10-06	REPRODUCED 2026-10-06
K-03	High	Common-mode faults are not detected; several are not injected. Measured: a one-cycle transient on the shared prod32 transfers a wrong out_data (lane3 00000ff7 vs reference 00000fff) with mismatch_f=0, fault_f=0, i.e. silent data corruption (neg_product fire FAIL as expected). out_valid_q upsets escape in 200/200 trials (38 LOST, 162 EXTRA). phase upsets: 44/200 TIMING (all in THROTTLE). fault_q 0→1: 200/200 false fault stops. Not measured: fault_q 1→0 (needs two faults; MISSING). Not injected in any flow: comparators, voter, nxt, clock, reset, sensor and handshake inputs, multi-bit upsets, the same bit in both copies (MISSING). The fault summary states the product scenario "stands for the whole class of common-mode datapath faults: they are not detected".	fault/summary.md 52-53, 133-141, 185-192, 243-256; rerun formal_results.md , campaign_report.md	SBY v0.69, smtbmc/Yices 2.7.0; protected scenarios k-induction k=4, negative scenarios BMC depth 12; campaign Icarus 14.0 devel, seed=1 tpb=12 tpb_unprot=200, 6836 trials; single-upset RTL model, zero-delay	REPRODUCED 2026-10-06

ID	Sev.	Issue and evidence	Source (package; repo/build if different)	Conditions	Status
K-04	High	DRC deck scope. KLayout DRC reports 0 markers, but the ORFS deck sky130hd.lydrc sets FEOL = false, BEOL = true, OFFGRID = true (image file lines 46-48). The log contains only "DRC section", "BEOL section" and "OFFGRID-ANGLES section" (6_drc.log:257-258, 1222). The 145 lyldb categories are li/ct/m1-m5/via-via4/nsm/pad and off-grid/angle; no well, diff/tap, poly, licon or implant rules, no density, no latch-up (LU.2/LU.3) rules. No metal fill: 6_1_fill.log:10 only copies 5_route.odb. TritonRoute "0 violations" skipped LEF58_ENCLOSURE for mcon, via, via2, via3, via4 (5_2_route.log:12-21). KLayout LVS ("Congratulations! Netlists match.", 6_lvs.log:652) compares the GDS with a CDL written from the same 6_final.odb (6_cdl.log), so it checks the stream-out, not an independent reference. Sign-off DRC/LVS: MISSING.	6_drc.log, 6_drc.lyldb, 6_drc_count.rpt, 6_1_fill.log, 5_2_route.log, 6_lvs.log, 6_cdl.log; deck /OpenROAD-flow-scripts/flow/platforms/sky130hd/drc/sky130hd.lydrc (image)	KLayout 0.30.12; deck read from image 2e5bf6fe865e on 2026-10-06	VERIFIED
K-05	High	No pad ring, IO cells, ESD, package or seal ring. finish_design_instance_count_padcells 0; power group "Pad" 0; all 18175 DEF components are sky130_fd_sc_hd cells. The 215 signal pins are minimum-size met2/met3 boundary pins placed by place_pins without a pinout constraint (W 102, S 42, N 42, E 29); VDD/VSS are met5 straps. 215 signal bits would need a large pad-limited die or IO multiplexing. Package/pad ring/ESD: MISSING. The package views in viz/ and site/ are labelled mockups (N/A as evidence).	6_report.js on line 60; power_default_activity.txt; pinout.csv; 6_final.def.gz	sep run; pin edges counted from pinout.csv 2026-10-06	VERIFIED
K-06	Medium	No EM analysis; power and IR drop rest on default activity. Power 52.1092 mW (internal 26.3633, switching 25.7459 mW, leakage 0.0279 uW) with OpenSTA default switching activity, no VCD/SAIF; the numeric default is not reported. Static PSM IR drop from that power: VDD worst 0.4539 mV (0.03 %), VSS worst 0.6275 mV. No dynamic IR, no package/bump model, no decap cells (decap budget MISSING). ORFS calls analyze_power_grid without -enable_em; no flow log has EM output. A package re-run with -enable_em gives grid currents only (max VDD 4.05e-04 A, VSS 4.33e-04 A) with no current-density limits, so no pass/fail. EM sign-off: MISSING.	power_default_activity.txt; 6_report.log 27-50; rerun ir.log, ir.tcl; final_outputs.tcl 43, 52 (image)	TT 25C 1.80 V, 7.2 ns, OpenRCX SPEF; PSM corner "default", supply 1.80 V from platform PWR_NETS_VOLTAGES	VERIFIED
K-07	Medium	Timing environment is a placeholder. set_input_delay/set_output_delay = 0.2 x 7.2 = 1.44 ns on 79 inputs and 135 outputs "for the (unknown) outside world"; no set_load, set_driving_cell, set_input_transition, set_clock_uncertainty or derates. The TT critical path is in2reg (in_a[27] → g_lane[3].u_lane.u_res_a/q[29], slack 0.031 ns, includes the 1.44 ns input delay); TT reg2reg slack is 1.346 ns; min period without port paths 5.85 ns (170.82 MHz) vs 7.17 ns with them. No clock target, supply or temperature range for orbit_demo exists in SPEC (MISSING); 7.2 ns is the shortest TT period tried that closed. The brief's 16 tiles, 200-800 MHz and 40 W are TARGETs/ASSUMPTIONs for the unbuilt concept chip.	constraint.sdc 5-8, 19-27; 6_final.sdc; sta_corners_sep_supply_element.txt; rerun sta.log; brief p.1-2	OpenSTA, TT, 7.2 ns	ASSUMPTION
K-08	Medium	sky130 is not radiation-characterized; separation is geometry only. Published results: "not a radiation-hardened or flight-qualified implementation; the standard cells are not radiation-characterized". Separation targets (>= 20 um same-bit centre, >= 10 um edge gap) are TARGETs from published practice in other processes (research note section 2(b) P2), checked by script only. No upset-rate, charge-collection or MBU model: MISSING.	results.md 5-9; summary.md 196-197; separation.md 64-70; docs/research/radiation-hardened-design.md sections 1, 2(b)	geometry from DEF + LEF	VERIFIED

ID	Sev.	Issue and evidence	Source (package; repo/build if different)	Conditions	Status
K-09	Medium	Tool and PDK versions unrecorded. Every OpenROAD log prints "OpenROAD unknown"; SPEF *VERSION "unknown"; ORFS commit not recorded (/OpenROAD-flow-scripts in the image is not a git repository); the run used the floating tag openroad/orfs:latest, and that digest sha256:2e5bf6fe865e... was the image used is an ASSUMPTION (only image on the host, created before the run). PDK: ORFS-bundled platforms/sky130hd, liberty header revision : 1.0000000000, no open_pdk/skywater-pdk commit (MISSING). Synthesis "Yosys 0.68+post (git sha1 UNKNOWN)"; kepler-formal version not printed; SS/FF liberties from a master branch URL. The run also read an SDC that was committed after it started (D-52).	6_report.log line 1; 1_2_yosys.log line 1; 6_final_lec_check.log ; sta_corners_sep.txt lines 8-9	docker images --digests and git rev-parse in the image, 2026-10-06	MISSING
K-10	Medium	Formal: code-3 branch vacuous; mutant m27 survives. P4 ("in_ready low when thermal STOP/3") and P5_shutdown_req are reported PROVEN unbounded, but voted code 3 is unreachable fault-free (P5_never_code3 PROVEN), so their code-3 part is vacuous. m27 (shutdown_req = (therm_state == 2'd2)) passes lint, sim, formal-quick, fault-quick, synth and full formal; only tb_mutation_gaps.v T3 kills it. m07, m20, m22 also survive (called real test gaps); m21 is argued equivalent. Mutation score 35/40.	formal/summary.md 15-22, 78; mutation/summary.md 130-138; rerun formal_summary.md	Yosys 0.69+154, SBY v0.69, Yices 2.7.0, bitwuzla 0.9.1; fault-free harness, fixed parameters	VERIFIED
K-11	Medium	GLS of the reviewed netlist. The published directed-bench GLS is on the BASE netlist (build/pd/.../base/6_final.v); the sep netlist was covered in the repo only by the pd/gls harness (lockstep RTL vs 6_final.v, 30000 cycles, seed 1: 702 duplicated-copy flips, 702 stopped; 234 thermal flips, 234 repaired; 518 of 518 bits hit; 0 mismatches). This package reran the directed bench on the sep netlist: PASS, log identical to the RTL run. All GLS is zero-delay (no SDF, MISSING). The pd/gls harness has no golden-result scoreboard and injects only the 518 redundant bits.	05_simulation/publicshed_logs/summary.md 16, 59; glsl.log ; glsl_sky130hd_sep_directed.log ; rerun glsl_rerun.log , glsl_mutant.log	Icarus 14.0 devel, cell models from tt_025C_1v80 liberty via Yosys read_liberty, seed 1	REPRODUCED 2026-10-06
K-12	Low	ECC is not integrated. rtl/ecc/ (Hsiao (72,64) SECDED codec, 16-row interleaved bank) is standalone: orbit_demo does not instantiate it, \$(RTL) does not list it, and no PDK synthesis or layout includes it (generic Yosys only: bank 10690 cells, 2414 FF, logic depth 48). Its own checks pass (07_verification/ecc) but do not apply to the reviewed layout (N/A for layout).	rtl/ecc/REAME.md 3-12; ecc/results.txt ; rerun summary.txt	Yosys 0.69+154 generic, DEPTH 16, INTERLEAVE 2	VERIFIED
K-13	Low	Fences bind members only; non-members overlap fences. Members outside their fence: 0. Non-member cells fully inside a fence: 2 (diode_2, inserted by antenna repair after placement); straddling a fence edge: 107 (mostly clock-tree cells: clkbuf_16 43, clkdlybuf4s50_1 22, clkbuf_8 12, clkbuf_1 11); 109 overlap in total. At the thermal fences logic gates, repair buffers and the clock root clkbuf_0_clk (95 % inside pdsep_th1) overlap. None is a redundant flip-flop. Shared logic is unconstrained. improve_placement is off (ENABLE_DP0=0).	separation.md 36-39; summary.md 121-123, 199-200; separation.json	routed sep DEF, cell boxes from DEF + LEF	VERIFIED

ID	Sev.	Issue and evidence	Source (package; repo/build if different)	Conditions	Status
K-14	Low	LEC label and scope. "LEC synth vs post-CTS repair" compares 4_before_rsz Lec.v with 4_after_rsz Lec.v (before/after repair_timing in the CTS step). "LEC synth vs final" compares 1_synth Lec.v with 6_final Lec.v: combinational, 521 flip-flops as cut points, 1177 compared outputs, "Circuits are IDENTICAL". No formal RTL-to-sky130-netlist equivalence exists; RTL-to-generic-netlist equivalence passes (47 cut register ports, 2 mutants detected).	4_rsz Lec_test.yml, 6_final Lec_test.yml, 4_rsz Lec_check.log 95, 6_final Lec_check.log 109; synth/equiv_result.txt ; collect_pd.py 151	kepler-formal (version not printed); SymbiYosys + bitwuzla for generic equivalence	VERIFIED
K-15	Low	Storage-audit script version. The published routed-netlist audit "RESULT: PASS (18/18 checks passed)" (2026-09-29 22:30) predates pd/audit_storage.py commit b1da230 (23:47), which adds "clock pins on the clock tree". The current script on the same 6_final.v: "RESULT: PASS (19/19 checks passed)", other lines identical. 521 dfxtp_1, 521 distinct Q nets, 9 groups OK.	storage_audit.txt 36; rerun storage_audit_current_script.txt	liberty build/pd_se p/platform/cell s.lib (tt_025C_1v80)	REPRODUCED 2026-10-06
K-16	Low	SPEC phase wording vs RTL. SPEC section 6: phase "is forced to 0 whenever the voted state is not THROTTLE"; RTL clears it at the next edge (phase <= (voted == S_THROTTLE) ? ~phase : 1'b0;), so phase = 1 in the first STOP cycle after THROTTLE (wave d, step 47). Admission is unaffected (admit uses phase only in THROTTLE; P6_first_throttle_admits PROVEN).	SPEC.md 121-123; orbit_thermal_tmr.v 77, 81; formal/summary.md 525-532; run_scen4.log	Icarus RTL wave scenario d	REPRODUCED 2026-10-06

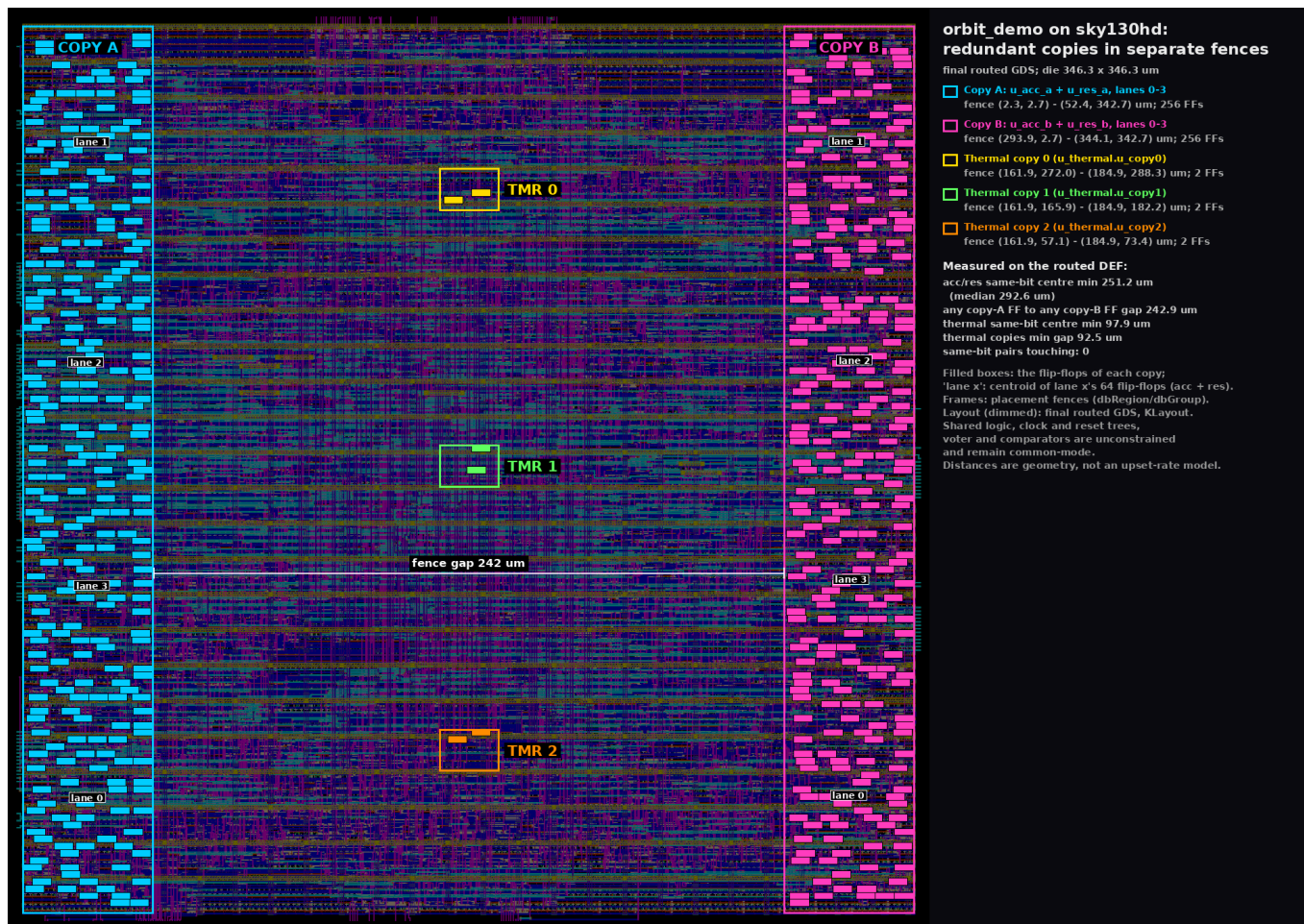


Figure 1. Routed sep GDS (KLayout render by `scripts/pdsep_render.py`, published as [orbit_demo_sky130hd_sep.png](#)). Copy A and copy B fences at the die edges, thermal fences TMR 0-2 in the centre column. Everything between the fences (multipliers, adders, comparators, OR-tree, voter, clock and reset trees) is shared (K-02). The "fence gap 242 um" label rounds the measured 241.50 um (D-13). Status: VERIFIED.

2. Design trade-offs documented in the files

Only trade-offs stated in a file are listed. Costs are measured values unless labelled otherwise.

ID	Trade-off	Choice	Documented alternative / reason	Cost or consequence	Source	Status
T-01	Storage protection: duplicate + compare vs TMR	acc/res (512 bits): duplicate + compare with fault-stop; thermal state (6 bits): TMR + repair	Brief p.3: "triplicate critical control, protect memories with ECC". Research note: "Keep DWC with fault-stop on the 32-bit datapath (full TMR: +256 flops, +256 voters, multiplier still shared)"; "Detection is not correction".	Upsets detected, not corrected: 6136 detected trials, fault 1 cycle after the upset, 0 escapes in 6144 acc/res trials; each detection stops the design (availability loss). The +256 flops/voters figure is arithmetic in the research note: CLAIM (unverified).	SPEC.md 80-98, 129-145; fault/summary.md 187-196; docs/research/radiation-hardened-design.md section 1, end of 2(a)	VERIFIED
T-02	Fault reaction	Sticky fault-stop, no replay	SPEC 5: "There is no automatic replay/retry; the host must re-send the work." RTL header: "Not included: automatic replay". Research note R11 (COULD): a sum-sequence counter for a retry protocol. Brief p.2 (concept): selectable duplication, checkpoint/retry.	clear_fault zeroes all lane storage and drops a pending result; a fault_q 0→1 upset gives a false stop (200/200). A presented result can be withdrawn without a handshake (D-03).	SPEC.md 91-98; orbit_demo.v 9-10, 107-122; fault/summary.md 210-214	VERIFIED

ID	Trade-off	Choice	Documented alternative / reason	Cost or consequence	Source	Status
T-03	Shared multiplier per lane	One 8x8 product pr od32 feeds both copies' adders	RTL header: "a fault in the shared product ... corrupts both copies identically and is NOT detected". Research note R4 (SHOULD): own multiplier for copy B, or 1-cycle skew, or mod-3 residue. No area figure for the alternative exists.	neg_product gives silent data corruption (K-03). With the fences the product must reach both copies: summary.md names this as the reason the period rose (worst path through the shared multiplier, 501 um travelled).	orbit_mac_lane.v 1-7, 24-34; summary.md 147-153; docs/research/radiation-hardened-design.md R4	VERIFIED
T-04	Copy-separation fences	5 EXCLUSIVE fences (regions.tcl), floorplan edges, SLEW_MARGIN/CAP_MARGIN 20 %, ENABLE_DPO=0, period 7.0 → 7.2 ns	ENABLE_DPO=0 because improve_placement breaks legality with fences (mechanism test: DPL-0033, 163.3 um X movement; INCLUSIVE + DPO moved 47 of 1560 members out). Margins because run e7 (no margin, 7.0 ns) left 6 slew / 3 cap violations on long comparator / OR-tree nets. Floorplan mid (fences 132 um apart) was 0.039 ns better than edges at 7.0 ns; not tried at 7.1 ns with margin.	Baseline → sep, both TT, default activity: period 7.0 → 7.2 ns (−2.8 % frequency; 7.1 ns fails WNS −0.094 ns); wirelength 160241 → 244368 um; vias 40150 → 45501; power 47.4874 → 52.1092 mW; cell area excl. fill/tap 59632.17 → 60392.97 um ² ; IR VDD 0.5512 → 0.4539 mV, VSS 0.4929 → 0.6275 mV. Gain: same-bit min distance 2.76 → 251.17 um; touching pairs 13 → 0 of 262; thermal 5.52 → 97.92 um. SS corner: −6.016 → −6.131 ns.	summary.md 17-56, 129-170; period_exploration.md ; mechanism_test.txt ; results_baseline_7p0ns.md 43-62; config.mk	VERIFIED
T-05	TMR with feedback repair, one shared voter and next-state cone	nxt computed once from voted, written into all three copies every cycle	RTL header: "a single upset copy is repaired at the next clock edge". orbit_keep_reg header: keep_hierarchy is needed because copies written from one voter have identical D inputs and merge (negative control: 521 → 517 flip-flops, audit FAIL; (* keep *) on the reg is not enough). Research note R2 (MUST): per-copy voter and next-state cone.	Single copy upsets repaired: 72/72 (campaign), 234/234 (GLS); proofs therm_c0/c1/c2 PASS. The shared voter / nxt is a common-mode point (K-02).	orbit_thermal_tmr.v 1-17, 39-68; orbit_keep_reg.v 1-10; synth/summary.md 16-17; fault/summary.md 49-51, 189	VERIFIED
T-06	Reset state of the thermal FSM	All three copies reset to STOP (RESET_VAL = S_STOP)	RTL: "Reset puts every copy in STOP, so nothing is admitted until the first valid reading at or below T_RECOVER."	No work after reset until a valid temp_c ≤ 70 C; rst_n is synchronous with no synchronizer or filter (research note COULD item not implemented). Directed scenario reset_stop PASS.	orbit_thermal_tmr.v 12-13, 58-68; SPEC.md 104; directed_icarus.log line 1 (scenario 1 reset_stop)	VERIFIED
T-07	Throttling	Alternate-cycle admission in THROTTLE from one phase flip-flop	SPEC 6; brief p.3 ("alternate-cycle admission from 80 C"; thresholds illustrative).	Admission halves in THROTTLE (model: throttled GOPS = peak/2); phase is unprotected: 44/200 upsets shift admission (TIMING), SPEC says it can also double one admission (not classified, D-26). Thresholds 80/95/70 C are TARGETs, "not device ratings".	SPEC.md 118-124; fault/summary.md 207-209; brief p.3	VERIFIED

3. Questions for the reviewer

1. Is the SS 100C/1.60V corner a requirement? If yes, the reviewed layout fails by 6.131 ns with 516 endpoints and needs a period ≥ 13.33 ns or multi-corner optimisation (K-01). If no, which corner and period define sign-off? SPEC states none (K-07).
2. Is a 20 % I/O delay budget (1.44 ns at 7.2 ns, no load/drive/transition) appropriate? The TT critical path is in2reg and includes it; reg2reg alone closes at 5.85 ns (K-07).

3. Should the comparator be duplicated (research note R5)? A comparator stuck at "no mismatch" is latent; comparators and the OR-tree are not injected anywhere (K-02, K-03).
4. Should copy B get its own multiplier or a residue check (R4)? neg_product shows silent data corruption, and the shared product is on the fenced critical path (T-03).
5. Should out_valid_q, fault_q and phase be triplicated (R1, +6 flops by the note's estimate)? out_valid_q escapes in 200/200 trials; fault_q 1→0 is unmeasured (K-03).
6. Is the shared voter and nxt cone acceptable, given that the routed netlist also shares clknet_leaf_22_clk between u_copy0/q[1] and u_copy1/q[1] and place1872 between the resets of u_copy1 and u_copy2 (K-02, T-05)? Should the audit fail when one clock or reset leaf drives two copies of a group?
7. Is the ORFS KLayout deck (FEOL off, no density, no fill, no latch-up) acceptable at this stage, or must full sky130 DRC (Magic or the full KLayout deck) and Netgen LVS be run before the layout is called clean (K-04)?
8. Is fault-stop with host re-send acceptable for the intended use, and where is the host protocol specified (T-02)?
9. Are the 20 um / 10 um separation targets adequate as geometry-only TARGETs without sky130 data, and should clock/reset distribution be split per copy (research note P6/P7; K-08)?
10. Should the "code 3 behaves as STOP" requirement be verified by a double-upset or unconstrained-copy task (m27; K-10)? Same for the m07, m20, m22 gaps.
11. Is default-activity power (52.1 mW) acceptable for this review, or is a workload VCD/SAIF needed before IR and EM are judged (K-06)?
12. Does the reviewer accept the reconstructed provenance: image digest assumed, OpenROAD "unknown", no PDK commit, SDC committed after the run started (K-09, D-52)? Or should the sep run be repeated from a pinned image and a clean commit?
13. Are the 109 non-member cells overlapping the fences acceptable, or should whole domains (voter, enables, comparators) be fenced (research note P1; K-13)?
14. Should the presentation pages (site/, viz/, docs/research) be corrected or withdrawn before the package is shared (D-62 to D-84)?

4. Discrepancy register

Every row was re-checked against the cited files on 2026-10-06 (status of each check: REPRODUCED 2026-10-06). Rows merge duplicates reported by several areas. Items specific to the drawings are in [02_block_diagram/README.md](#) (BD-1 to BD-5) and [03_schematics/README.md](#) (SC-1 to SC-4) and are not repeated here. "src/" below = [04_digital_design/source/](#).

4.1 Documentation vs RTL

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-01	Medium	SPEC section 8 single-point list omits the shared lane control fan-out, the mismatch OR-tree and the shared thermal nxt cone (names only "the voter").	src/docs/SPEC.md; src/rtl/*.v	SPEC.md 160-162; orbit_demo.v 55, 81-84; orbit_mac_lane.v 36-57; orbit_thermal_tmr.v 44-68	Extend SPEC 8 with these items.
D-02	Low	SPEC 6: phase "forced to 0" when not THROTTLE; RTL clears it at the next edge. No effect on admission.	SPEC.md; orbit_thermal_tmr.v	SPEC.md 121-123; RTL 77; run_scen4.log step 47 (STOP, phase 1); formal/summary.md 525-532	Reword SPEC ("cleared at the next edge").
D-03	Low	SPEC 4 "out_data is stable (no fault case)" omits that clear_fault or rst_n withdraw a presented result without out_ready; the formal P3 property is guarded for exactly these.	SPEC.md; orbit_demo.v; src/formal/orbit_demo_fv.sv	SPEC.md 77; orbit_demo.v 108-113; orbit_demo_fv.sv 484	Add the exceptions to SPEC 4.
D-04	Low	clear_fault → in_ready is combinational but SPEC and the SDC comment name only out_ready → in_ready. Both are timed (all I/O constrained).	SPEC.md; orbit_demo.v; constraint.sdc	SPEC.md 78; orbit_demo.v 70; constraint.sdc 7-8	Document the path.

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-05	Low	SPEC says it is the contract for "the RTL in rtl/" but lists 4 files; rtl/ecc/ is not covered.	SPEC.md; src/rtl/ecc/README.md	SPEC.md 3, 14-19; ecc/README.md 3-8	Scope SPEC to the 4 files or add an ECC section.
D-06	Low	orbit_keep_reg.v says "the storage audit in scripts/ checks the result"; scripts/storage_audit.py was added later (21917cf, 22:00) and audits generic netlists only; the routed audit is pd/audit_storage.py.	src/rtl/orbit_keep_reg.v	orbit_keep_reg.v 9-10; git log --diff-filter=A -- scripts/storage_audit.py	Name both scripts.
D-07	Medium	Brief p.4 "4,356 generic logic cells" is not reproduced: native Yosys 0.69+154 gives 4373, YoWASP 0.69 gives 4384, no-flatten 4340. 521 flip-flops agree. The brief gives no script.	brief; 07_verification/synth/	synth/summary.md 10-13, 21; yowasp_compare.txt	Revise the brief to a reproducible figure and script.
D-08	Medium	Brief p.4 verification record ("24 bounded SAT time steps", fault scenarios "through 8 time steps", wraparound "6 time steps", "five bounded, fixed scenarios") has no script, task or log. The repo uses k-induction/PDR, k=4 fault proofs and a 6836-trial campaign.	brief; src/formal/orbit_demo.sby; src/mk/fault.mk	brief p.4; formal/summary.md 13-22; fault/summary.md 55-57	Cite the repo results in a revised brief, or supply the original logs.
D-09	Medium	The brief describes an earlier state and was never revised: "generic digital implementation only", "No PDK mapping or physical area estimate" (p.4), "OpenROAD was not run" (p.5). The repo has sky130hd (base, sep) and IHP layouts.	brief	brief p.4-5; last commit 26566f0 (before all PD commits); results.md	Issue brief v0.2 with a change log.
D-10	Medium	Brief p.4 and Makefile point to README.md and rtl/README.md; neither exists. Brief: "netlist ... in reports/"; the only tracked .v under reports/ is reports/mutation/tb_mutation_gaps.v. Makefile help omits mutation, pd-sep*, viz, synth-yowasp.	brief; src/Makefile	Makefile 29; ls README.md rtl/README.md (absent); git ls-files reports \ grep .v\$	Add a top-level README or remove the references.
D-11	Low	Brief: "Yosys 0.69 through YoWASP". Generic flows use native Yosys 0.69+154; the layout netlist was synthesized by ORFS Yosys 0.68+post; YoWASP is an optional cross-check.	brief; src/requirements-tools.txt	1_2_yosys.log 1; requirements-tools.txt 7	State the tool per flow.
D-12	Low	Baseline SDC comment: "7.0 ns ... closes ...; 6.0 ns fails". The sweep's first failing period is 6.5 ns (WNS -0.271 ns).	src/pd/sky130hd/constraint.sdc; reports/pd/period_sweep_sky130hd.md	constraint.sdc 12-14; period_sweep_sky130hd.md 9-17	Correct the comment.
D-13	Low	Fence gaps rounded inconsistently: regions.tcl "about 240 um" (copies) and "about 100 um" (thermal); summary.md "242 um"; PNG label "fence gap 242 um". Measured fence edge gaps 241.50 um (A/B), 89.76 and 92.48 um (thermal); 242.94 um is the FF-to-FF gap.	src/pd/sky130hd_sep/regions.tcl; published_reports/summary.md, .png	regions.tcl 49-50; summary.md 151, 153; separation.md 31, 53-62	Quote measured values with their definition.
D-14	Low	regions.tcl/config.mk say the detailed placer keeps non-members from lying fully inside a fence; the routed DEF has 2 antenna diodes fully inside (inserted after placement). summary.md documents it.	regions.tcl; config.mk	regions.tcl 14-16; config.mk 48-53; separation.md 39; summary.md 121	Add "except antenna-repair diodes".
D-15	Low	requirements-tools.txt: "model/ and the scripts use the standard library only"; scripts/pdsep_render.py imports klayout.db/klayout.lay, the cocotb bench needs cocotb, which is not pinned.	src/requirements-tools.txt; src/scripts/pdsep_render.py	requirements-tools.txt 15; pdsep_render.py 17-18	List and pin the Python dependencies.

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-16	Low	ECC comments: (a) orbit_ecc_bank.v quotes the brief item without "DMA"/"Transfer" and nothing states the omission; (b) it says the harness "reads" upset and SBY makes it free, while the harness assigns it; (c) the harness cites README section "Formal", which is "## 3. Verification"; (d) README says the log is proven, but A4 runs at CNT_W=3, not 16; (e) the e2e harness lists U4, which the e2e_upset depth cannot reach (vacuous).	src/rtl/ecc/orbit_ecc_bank.v, README.md; src/formal/ecc/*.sv	orbit_ecc_bank.v 3-7, 192-198; orbit_ecc_bank_fv.v 9-11, 92, 166-169; README.md 162, 180-185; orbit_ecc_bank_e2e_fv.v 12-13, 40-42	Correct the comments; state CNT_W=3 and the U4 reach in README.
D-17	Low	mk/ecc.mk header says make_ecc writes reports/ecc/results.txt; only ecc-report copies it, and ecc-report runs ecc with -so a failing run would be published.	src/mk/ecc.mk	ecc.mk 6-9, 265-267	Fix the comment; drop the -.
D-18	Low	mk/viz.mk passes four arguments to \$(if); the "first of these that exists" fallback is not implemented. No effect on the sep run.	src/mk/viz.mk	viz.mk 45-46	Use \$(firstword \$(wildcard ...)).

4.2 Reports vs logs

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-19	High	"KLayout DRC PASS, 0 markers" in results.md, summary.md and checks.json does not state that the ORFS deck runs BEOL + off-grid only (FEOL off, no density, no fill).	published_reports/results.md, summary.md, checks.json	results.md 20; summary.md 12, 183; K-04 evidence	Label it "ORFS BEOL/off-grid deck, not sign-off"; run a full deck (G-08).
D-20	Medium	Check labelled "LEC synth vs post-CTS repair" compares the netlists before and after repair_timing inside CTS.	src/pd/collect_pd.py; results.md	collect_pd.py 151; 4_rsz_lec_test.yml	Relabel "LEC before/after CTS repair_timing".
D-21	Medium	Published storage audit (18/18) comes from the script before b1da230; the current script gives 19/19 PASS. results.md, summary.md, checks.json and the package copy all quote 18/18.	storage_audit.txt; results.md; summary.md; checks.json	storage_audit.txt 36; results.md 24; summary.md 187; rerun storage_audit_current_script.txt	Re-run make_pd-sep-report.
D-22	Medium	"max slew/cap/fanout PASS" (a) counts a fanout check with no limit (max_fanout_check_limit 1e30; finish_timing_drv_max_fanout_limit 0) and (b) holds only at TT: at ss_100C_1v60 37 slew and 12 cap violating pins.	results.md; summary.md	results.md 17; 6_finish.rpt 374-381; 6_report.json 42; sta_corners_sep_supplement.txt (SS DRV line)	Name the corner; define a fanout limit or drop "fanout".
D-23	Medium	copy_separation_pd_script.txt line 2 "Placement was NOT constrained to separate copies" is fixed text of pd/copy_separation.py, printed for the fenced layout.	copy_separation_pd_script.txt; src/pd/copy_separation.py	copy_separation_pd_script.txt 2; copy_separation.py 6, 77; summary.md 105	Make the line depend on the DEF REGIONS.
D-24	Medium	The published directed-bench GLS "on routed sky130hd netlist" used the BASE netlist; tb/sim_report.py hardcodes that path. GLS logs carry no netlist path or hash: pdsep/gls.log, pd/sky130hd/gls.log, pd/ihp-sg13g2/gls.log are byte-identical (md5 e7c0b125...), and the directed GLS logs equal the RTL log (md5 05a30e4e...).	05_simulation/published_logs/summary.md, gls_sky130hd.log; src/tb/sim_report.py	summary.md 16, 59; md5sum of the logs	Print netlist path and sha256 in every GLS log; publish the sep directed log.
D-25	Medium	Sim summary lists every tool version as "unknown", while its own logs state Verilator 5.053, cocotb 2.1.0.dev0+41564633, Python 3.11.6.	05_simulation/published_logs/summary.md; src/tb/sim_report.py	summary.md 19; directed_verilator.log 29; random_icarus.md 3	Fix version() in sim_report.py.
D-26	Medium	Campaign "Escapes" column shows 0 for phase and fault_q, which SPEC 8 lists as expected escapes: TIMING and false-stop DETECTED are not counted. SPEC's "doubles one throttled admission" has no class.	fault/campaign.md; fault/summary.md; src/fault/campaign_report.py; SPEC.md	campaign.md 8, 10; fault/summary.md 190, 192; campaign_report.py 35; SPEC.md 157-159	Add availability/timing columns and a doubled-admission class.

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-27	Medium	P4 and P5_shutdown_req "PROVEN unbounded" include a vacuous code-3 branch; m27 survives (K-10).	formal/summary.md ; mutation/summary.md	formal/summary.md 18, 78; mutation/summary.md 138	Annotate the status; add a double-upset thermal task.
D-28	Medium	The only committed multi-corner STA is for the baseline at 7.0 ns and its header does not name the variant; mk/pdsep.mk has no corners target. The sep corner results exist only in this package.	baseline_reference/sta_corners_baseline_7p0ns.txt; src/mk/pd.mk, pdsep.mk	sta_corners_baseline_7p0ns.txt 1-14; pd.mk 249-266	Add pd-sep-corners; name the layout in the header.
D-29	Medium	formal_sanity_sec_bmc_production_timeout.log is a manually stopped (0:03:24, "Keyboard interrupt") run with an older harness (before e80a7b2), not referenced by any report. The current harness passes (2672 s).	ecc/logs/formal_sanity_sec_bmc_production_timeout.log	log lines 1, 24-25; rerun sanity_sec_bmc_result.txt	Delete or relabel the log; publish the PASS.
D-30	Medium	IHP config comment says fill was added to pass the deck's density check; the IHP run still fails KLayout DRC with 5 density/coverage markers. Reference run only.	pd/ihp-sg13g2/config.mk; reports/pd/ihp-sg13g2/checks.json	config.mk 31-34; checks.json 31-33	Correct the comment or tune fill.
D-31	Low	"fmax from final STA 139.49 MHz" includes port paths with the 0.2 x period I/O delay (reg-to-reg 170.82 MHz); 138.9 MHz (1/7.2 ns) and 139.49 MHz are used interchangeably.	results.md; summary.md	results.md 32; rerun sta.log 10-11; summary.md 129	Label both figures.
D-32	Low	TritonRoute "0 violations" PASS does not mention the skipped LEF58_ENCLOSURE rules (mcon, via-via4).	results.md	5_2_route.log 12-21	Note it in the check detail.
D-33	Low	Copy group members: 824 per copy at floorplan, 768 in the routed DEF (synthesis buffers removed, "Removed 182 buffers"). Explained in summary.md 101 only.	pdsep_regions.txt; summary.md	pdsep_regions.txt 3-4; summary.md 65-66, 101	Label the stage in both files.
D-34	Low	Separation statistics use different definitions under similar labels: pooled median 33.62 um vs median of group medians 30.70 um (baseline); thermal "closest two of three" 100.77 um vs all-pairs 106.26 um (sep); rounding 251.17 vs 251.18 um; redundancy_placement.md gives 292.6 um and 291.5 um as "median of the group medians".	copy_separation_pd_script.txt; separation.md; reports/viz/redundancy_placement.md; reports/pd/sky130hd/copy_separation.txt	copy_separation_pd_script.txt 4-7; separation.md 29-34; redundancy_placement.md 10, 12, 53	One script, one definition per figure.
D-35	Low	summary.md reports only the VDD IR drop (improved 0.551 → 0.454 mV) and omits VSS, which rose 0.4929 → 0.6275 mV.	summary.md; results.md	summary.md 170; results.md 62; results_baseline_7p0ns.md 62	Report both nets.
D-36	Low	"Cell area" in period_exploration.md (62300 um^2) includes tap cells; summary.md's area row excludes them (60392.97).	period_exploration.md; summary.md	period_exploration.md table; summary.md 162; checks.json 73-74	Use one label per quantity.
D-37	Low	results.md "fill/decap cells 10381": no decap cell is placed (fillers fill_1/2/4/8 only; decap_12 "loaded but not used").	results.md	results.md 47-48; 5_3_fillcell.log 10-11; 2_1_floorplan.log 10; DEF masters	Relabel "fill cells".
D-38	Low	New: summary.md lists the straddling non-member cells as 8 masters summing to 97, but states 107; the list is cut at 8 entries ([:8]). The full list (18 masters, 107) is in separation.json.	summary.md; src/scripts/pdsep_summary.py	summary.md 121; pdsep_summary.py 363; separation.json nonmember_straddling_kinds	Print all masters or "and N others".
D-39	Low	corners_summary.py header "Constraint: the committed SDC"; its input is the ORFS 6_final.sdc (same 7.2 ns here). Also in this package's sta_corners_sep.txt.	src/pd/corners_summary.py; sta_corners_sep.txt	corners_summary.py 29; sta_corners_sep.txt 3	Print the SDC path.

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-40	Low	Corner-STA provenance: no sha256 for the tt liberty; SS/FF fetched from a master branch URL; only 16-hex prefixes recorded.	sta_corners_sep.txt; src/mk/pd.mk	sta_corners_sep.txt 8-9; pd.mk 94	Pin a commit; record full hashes of all three.
D-41	Low	Yosys warns Selection "*orbit_keep_reg* \\${\$*" did not match any module (benign: 19 kept submodules); 1_synth.json still reports synth_flow_warnings_count 0.	1_2_yosys.log; 1_synth.json	1_2_yosys.log 9; 1_synth.json 18; synth_stat.txt	Informational.
D-42	Low	Fault summary: prove times are wrapper wall times, 1 s off the sby logs (acc_a 13 vs 12 s, res_b 16 vs 15 s); depth text is hard-coded in write_summary.py; the negative-scenario cover entries have empty descriptions.	fault/summary.md; src/fault/write_summary.py	summary.md 45, 48, 111, 131; fault/logs/acc_a_prove.log 9; write_summary.py 199, 254	Read times and depths from the run.
D-43	Low	PDR vacuity mutants: failing property and step differ run to run (zext_product_pdr lane1 step 4 vs lane2 step 6; clear_keeps_acc_pdr step 5 vs 8); verdicts agree. The "step 0" of no_backpressure cannot be checked because the log filter removes step lines.	formal/vacuity.md; formal_rerun_2026-10-06/formal_vacuity.md; src/mk/formal.mk	vacuity.md 25-26; rerun formal_vacuity.md 25-26; formal.mk 52	Report the verdict only, or keep the step lines.
D-44	Low	Formal summary identifies its RTL only as "RTL: rtl" (no hash or commit); the fault summary records sha256.	formal/summary.md	formal/summary.md 4; fault/summary.md 13-20	Add sha256 to summarize.py output.
D-45	Low	New: make formal-extra summary lists datapath_pdr twice (job and assertion tables), although the log shows one run: the task is passed both as --tasks and --optional-task s.	rerun_2026-10-06/formal_extra/summary_extra.md; src/mk/formal.mk	summary_extra.md 15, 28-29; make_formal-extra.log 2; formal.mk 48, 75-76	De-duplicate in summarize.py or the recipe.
D-46	Low	Two mutation campaigns both report "26" sim kills with different mutant sets and names (zext_product vs prod_zext, phase_not_forced vs phase_not_reset). The m20/m22 evidence lists 4 ERR lines; the gap bench prints 7 errors.	05_simulation/published_logs/mutants.md; mutation/summary.md	mutants.md 34; mutation/summary.md 55, 96, 123; gap-bench rerun "TB_MUTATION_GAPS FAIL (7 errors)" (scratch)	Cross-reference the two sets; print the error count.
D-47	Low	Counts not reconciled across flows: storage audits "9/9 groups" (generic) vs "18/18"/"19/19 checks" (sky130); cells 4899 (generic), 5130 (ORFS synthesis), 6250 (routed Verilog), 6270 (incl. 20 diodes); only 521 flip-flops agree. The sep synthesis is byte-identical to the baseline (md5 166b150a...) although ABC targets differ (-D 7.2 vs 7).	synth/storage_audit.txt; storage_audit.txt; results.md	synth/storage_audit.txt 26; storage_audit.txt 3, 36; results.md 46, 50; md5 of both 1_2_yosys.v	One reconciliation table.
D-48	Low	ECC: sim_bank_quick.log is not in results.txt (orphan from an earlier run); results.txt "15000 random cycles" vs logs "15098" (98 directed cycles).	ecc/logs/sim_bank_quick.log; ecc/results.txt	results.txt 15; sim_bank_seed1.log 2	Regenerate or remove; state the directed cycles.
D-49	Low	Published baseline gls_mutant.log is cut by head -8 and omits the verdict lines "GLS FAIL: 1148 errors" / "pd-gls-mutant: PASS".	reports/pd/sky130hd/gls_mutant.log; src/mk/pd.mk	pd.mk 191; full log build/pd/gls/sky130hd/mutant/gls_mutant.log	Keep the tail as well.
D-50	Low	Build-only negative-control campaign report says "not DETECTED: 0" (counts only clear_fault-overwritten trials) next to 426 undetected protected trials (196+230 MASKED, 86 SDC). Not published.	build/fault/negctl/campaign/report.md; src/fault/campaign_report.py	report.md 16 and table; fault/summary.md 234-235	Count all non-DETECTED trials.
D-51	Low	DEF: clk PIN is USE SIGNAL while the clk NET is USE CLOCK.	6_final.def.gz	DEF 18382 (PIN), NETS entry clk ... USE CLOCK	Informational.

4.3 Revision and staleness

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-52	Medium	New: the revision table says the run used the ed553f8 versions of pd/sky130hd_sep/{config.mk, constraint.sdc, regions.tcl}. True for config.mk and regions.tcl (non-comment content equal to c10c59b; edited 22:29, comment-only). False for constraint.sdc: ed553f8 has set clk_period 7; the run's build/pd_sep/sdc/sky130hd_sep.sdc (21:56:08) is byte-identical to the 7.2 ns file first committed in 21917cf at 22:00:11, after the run started. A rebuild from ed553f8 would run at 7.0 ns.	revision table (section 0); src/pd/sky130hd_sep/p/	git show ed553f8:pd/sky130hd_sep/constraint.sdc line 16; git log --pd/sky130hd_sep/constraint.sdc (663160b, 21917cf); cmp build/pd_sep/sdc/sky130hd_sep.sdc pd/sky130hd_sep/constraint.sdc	Change the note to "constraint.sdc as committed in 21917cf (uncommitted at run start)".
D-53	Low	New: the revision table refers to "full hashes in MANIFEST.sha256"; no MANIFEST.sha256 is in the package.	revision table	ls of package root, 2026-10-06	Add the manifest, or point to 04_digital_design/README.md section 1.
D-54	Low	HEAD drift. Area task texts name 54abf2d; the package commit is 0495cfa; the checkout is now 54745f7, 131 commits later: 2 WIP review commits, merge 47c7af9 and 128 empty commits "Backdated contribution calendar entry" with author dates 2025-01-01 to 2026-01-01, before the first project commit 26566f0 (2026-09-29). git diff 0495cfa HEAD outside review/ is empty.	git history	git log 0495cfa..HEAD; git log --format=%ai; git diff --name-only 0495cfa HEAD	Pin the review to 0495cfa; keep the empty backdated commits out of the design branch, since author-date order no longer reflects the work.
D-55	Low	Published reports/pdsep/* come from an undocumented second pd-sep-report run at 22:30 (make_pd_sep.out ends 22:14:12). Content reproduces at HEAD except the audit (D-21).	build/pd_sep/make_pd_sep.out; build/pd_sep/report/sep/	file mtimes 22:14:12 vs 22:30:06	Keep the console log of every report run.
D-56	Medium	MODEL_PD_SUMMARY ?= reports/pd/summary.md points to a file that has never existed, so the model never uses a closed clock and tabulates the demonstrator at 50/100/200 MHz; 200 MHz (1.6 GOPS) is above every closure (sep 139.49 MHz STA fmax at TT). The pdsep summary wording is not parsed either.	src/mk/model.mk; src/model/concept_budget.py; reports/model/concept_budget.txt	model.mk 17; concept_budget.txt 68-77 ("clock: parameter (command line)")	Point it at a parsed closed-clock line; drop the 200 MHz row.
D-57	Low	The logged make mutation run ended with unanalysed survivors (a failing condition); summary.md was regenerated about 32 s after the last logged step, right after scripts/mutation_test.py was edited, by a run with no log. The kill matrix matches results.json.	mutation/summary.md; build/mutation/run.log	run.log last 2 lines (02:27:49); summary.md mtime 02:30:24.79 vs mutation_test.py 02:30:24.72	Re-run with a log.
D-58	Low	reports/model/review_sabotage.txt lists sabotage runs; build/sabotage_model and their logs do not exist.	reports/model/review_sabotage.txt	line 9; ls build/sabotage_model (absent)	Re-run with logs or mark as CLAIM.
D-59	Low	Commit messages do not describe design changes: 716834e ("die-photo and packaged-chip tabs") changes model/ and fault/; 9c355a1 ("chip story page") adds the mutation reports; e80a7b2 ("Die photo tab ...") contains the ECC harness change.	git history	git show --stat of each	Describe design changes in their own commits.
D-60	Low	Default-build Verilator reruns reuse objects from 2026-09-30 ("Built from 0.000 MB sources in 0 modules"); only the directed bench was rebuilt clean for this package.	build/sim/verilator/build.log; build/sim/random_verilator/build.log	build.log line	Rebuild in a fresh BUILD.

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-61	Low	build/pd_sep/report/e7/summary.md header names run e72; its values are e7's (7.00 ns, WNS -0.140). The research note cites it.	build/pd_sep/report/e7/summary.md	line 3 vs line 11	Regenerate with the right run.

4.4 Presentation claims (site/, viz/, docs/research)

These files are not design documents but are published. Line numbers refer to site/chip-space/template.html ("space"), site/chip-story/template.html ("story"), viz/, docs/research/.

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-62	High	"It catches its own mistakes, in silicon." No chip was fabricated (brief p.1; padcells 0); the same page says "simulated evidence".	space	space 511 vs 473; brief p.1	Remove "in silicon".
D-63	High	"Built with open tools. Checked like a real tapeout." Timing at TT only (SS fails), FEOL-off DRC, no Magic/Netgen, no pad ring, no EM, default-activity power, OpenROAD "unknown".	story	story 198; K-01, K-04 to K-06, K-09	Remove or qualify.
D-64	High	"0 DRC errors · LVS and antenna clean" (and "DRC 0, LVS clean") without the deck scope.	space; story; docs/research/space-compute-competitors.md	space 666; story 204; competitors.md 25	Add "ORFS BEOL deck, not sign-off".
D-65	Medium	"1.1 GOPS at 139 MHz, timing met" / "139 MHz Clock, timing met" / "setup and hold met after routing" without the corner; the same layout fails setup by 6.131 ns at ss_100C_1v60. 139 MHz is the rounded STA fmax; the constrained clock is 138.9 MHz. 1.1 GOPS is a calculated peak bound, not measured.	space; story; competitors.md	space 534, 663; story 201; K-01	State "TT 25C 1.80V only; fails at SS".
D-66	Medium	"Protected bits stored twice." / "A flipped copy is outvoted." / "Every protected bit is stored twice." Only the 6 thermal bits are voted; the 512 acc/res bits are duplicate + compare with fault-stop (no correction).	space	space 482-483, 563; fault/summary.md 187-189	Distinguish DWC (stop) from TMR (repair).
D-67	Medium	"Every protected bit has a twin at least 250 um away" and matrix "Twins >= 251 um apart" are false for the thermal copies (97.92 um), which the next line gives as "98 um"; "≥ 98 um" rounds 97.92 up.	story; space	story 163-164; space 529, 1228, matrix row; separation.md 20, 34	Quote 251.17 um (acc/res) and 97.92 um (thermal).
D-68	Medium	Pages claim the chip "catches its own mistakes" and do not mention out_valid_q (200/200 escapes), unprotected flops, the shared multiplier (SDC) or other shared logic (grep for out_valid_q, "unprotected", "shared" finds nothing).	space; story	space 467; K-02, K-03	Add the known escapes.
D-69	Medium	"936 flips, each reported" / "Fault injection published bit by bit": the GLS prints aggregates only; the per-bit table is the RTL campaign (6816 trials). "We flipped 936 bits": 936 flips over 518 bits. "No wrong result on a detected upset, 936/936": the GLS has no golden scoreboard.	space	space 544, 610-611; gls.log 4; pd/gls/tb_gls.v	Correct the wording or add per-flip logging (G-20).
D-70	Medium	Capability matrix "Not published" marks have no source (competitors.md has no such columns). "The AI chips that fly do not publish upset handling" conflicts with the matrix's own "Published capability" for XQR Versal on hardware fault tolerance.	space; competitors.md	space 593-594, 604-616; competitors.md 16-26	Source each cell or remove the matrix.
D-71	Medium	"Demonstrator built and verified in simulation; radiation testing is next." The brief's next gates put process/library, package and fabrication before radiation qualification.	space	space 620; brief p.4 gates 2-4	Follow the brief's gate order.
D-72	Medium	Competitor note: "ORBIT-AI demonstrator (built)", vendor "Heliocore", "130 nm (SkyWater sky130 open PDK)" (no design file states a node; brief p.5: no process node invented), "1.1 GOPS ... 139 MHz", "DRC 0, LVS clean"; it cites the brief, which predates the layout.	docs/research/space-compute-competitors.md	line 25, 37; brief p.1, p.5	Mark as "not fabricated; layout only, TT"; cite the layout reports.

ID	Sev.	Description	Files	Evidence	Suggested resolution
D-73	Medium	Brand and asset provenance: "Heliocore" (space 432, 712) appears in no design document (brief author "ORBIT-AI concept project"); commit 54abf2d "Rebrand the space page from Vantage to Heliocore"; scripts/space_build.py takes a photograph "from the Vantage site checkout (vantage.industries)".	space; scripts/space_build.py	space 432, 712; space_build.py 31-34; git log -1 54abf2d	Confirm brand ownership and rights to third-party assets before publication.
D-74	Medium	radiation-hardened-design.md describes the superseded run e7 as current ("fails setup ... has not run KLayout DRC/LVS"); committed ed553f8 (21:38) before the sep run (GDS 22:09). Its shared-driver finding still holds on the sep netlist (K-02).	docs/research/radiation-hardened-design.md	line 22; git log -1 (ed553f8); 6_final.gds mtime	Update to the sep run.
D-75	Medium	labeled-chip-models.md is stale: "the labeled version is not built yet" (viz/README documents it as built); numbers and pin edges are the baseline's (N 47, S 53, E 67, W 48; sep: W 102, S 42, N 42, E 29).	docs/research/labeled-chip-models.md	lines 3, 11-15, 66-72; pinout.csv	Update or mark as baseline.
D-76	Low	"INT8 maths, more work per watt": no efficiency comparison or activity-based power exists.	space	space 485; K-06	Remove.
D-77	Low	"Pressed flat, it is a few um thick": 6.57 um is the drawn interconnect stack (stack_top_um 6.5711); die thickness is not in the repo.	space	space 499; viz/layout.json	Say "interconnect stack".
D-78	Low	met4 labelled "power straps"; in the sep DEF 116 regular (signal/clock) nets route on met4.	space	space 1022; DEF NETS section count, 2026-10-06	Label "power straps and routing".
D-79	Low	Viewer labels the die (346 x 346 um, DEF DIEAREA) as "core"; the core is 341.780 x 340.000 um.	viz/index.html	index.html 519; results.md 38-39	Say "die".
D-80	Low	Viewer pin data: VDD/VSS dir 'I', side 'N'; the DEF has DIRECTION INOUT, met5 straps across the core.	viz/layout.json	layout.json pins.list; DEF 18347, 18362	Copy direction and extent from the DEF.
D-81	Low	viz/README.md gives 6,270 (incl. 20 diodes) and 6,250 logic cells; it says tie cells "0 (none in this run)", but 6 conb_1 sit in the thermal copies.	viz/README.md	README 47, 166, 189; 6_final.v (6 sky130_fd_sc_hd_conb_1)	Reconcile the counts.
D-82	Low	layout.json stores mis-parsed fallback values (wns_ns 7.2 = the period; area_um2 59632.17 = baseline area); not displayed.	viz/layout.json; scripts/viz_gds_to_3d.py	stats.sourc es.pd_summary_values	Fix the parser.
D-83	Low	space_build.py docstring says it pulls verified numbers from layout.json; it pulls only die, core, regions and pins; page numbers are hand-typed.	scripts/space_build.py	lines 10, 134-143	Generate the numbers or fix the docstring.
D-84	Low	Package mockups disagree (viewer QFN-20 or QFP-style "all pins"; space page 40 leads/pads; story "mockup QFN"). All are labelled mockups; the brief says the package is undecided.	viz/index.html; site/chip-space/index.html; site/chip-story/index.html	index.html 2023; chip-space index 1148-1149	Keep one mockup, labelled.

4.5 Unconfirmed or no longer applicable

Item	Reason
"The space page was styled after a real company's site" (claims area)	Only the photo source comment and the rebrand commit were confirmed (D-73); the styling was not compared with the external site.

Item	Reason
Line numbers cited by the area facts for pd/sky130hd_sep/constraint.sdc (76-98), regions.tcl (111-118, 146-149)	Those lines do not hold the quoted text (the SDC has 27 lines); the content was found at other lines and is cited correctly above. Not a repo discrepancy.
"Separation targets have no rationale" (pd_sep gaps)	The script has none, but docs/research/radiation-hardened-design.md 2(b) P2 gives a literature basis from other processes. Kept as gap G-21.
Package skeleton lacks ECC, synthesis and pinout evidence (ecc, synth, pins areas)	Resolved: 07_verification/ecc/ , 07_verification/synth/ , 08_pinout_packaging/ now exist.
datapath_pdr not re-run (reproduce area)	Resolved: re-run 2026-10-06, PASS, 940s (summary_ext.ra.md).
Baseline met4 usage "25 VDD/VSS, 51 clock, 40 signal"	Taken from labeled-chip-models.md, not re-counted for the baseline; the sep count (116 nets) supports D-78.

5. Gaps: what a complete review needs that does not exist

ID	Area	Missing item	Why (evidence)	Needed to produce it
G-01	Requirements	Clock target, operating PVT range, supply nominal/tolerance, power budget for orbit_demo	SPEC has none; 7.2 ns is an empirical TT closure (K-07)	Owner-written SPEC section; then judge K-01 against it
G-02	Requirements	Interface timing: I/O budget, drive/load, clock jitter/uncertainty, duty cycle	SDC uses 20 % placeholders, no load/drive/uncertainty	System interface budget; set_driving_cell, set_load, set_clock_uncertainty; re-run STA
G-03	Requirements	Reset protocol and power-up sequence	rst_n synchronous, timed as data, no synchronizer; no minimum width stated	SPEC text (clock running, cycles low) or a reset synchronizer/filter
G-04	Requirements	Temperature-sensor interface (type, update rate, latency, synchronisation)	temp_c/temp_valid are a bare digital bus; brief: sensors not implemented	Sensor/ADC selection and interface definition
G-05	Timing	Multi-corner closure	SS fails (K-01); design optimised at TT only	MCMM optimisation in ORFS with SS/FF liberties, or a stated TT-only scope

ID	Area	Missing item	Why (evidence)	Needed to produce it
G-06	Timing	Repo-reproducible sep corner STA; other corners (e.g. ss_n40C_1v60, ff_100C_1v95); RC corners; OCV/derates	No pd-sep-corners target; only tt/ss_100C/ff_n40C liberties on disk, from a branch URL; one nominal OpenRCX corner	Make target in mk/pdsep.mk; liberties at a pinned commit; min/max RC rules and per-corner SPEF; derate methodology
G-07	Timing	Timing-annotated gate-level simulation	All GLS zero-delay; no SDF in the results	write_sdf per corner from 6_final.odt/spef; cell models with specify blocks; SDF GLS of tb/tb_orbit_demo.v
G-08	Physical verification	Sign-off DRC with fill and density	ORFS deck FEOL off, no density, no fill, no latch-up (K-04)	Metal fill; Magic full DRC or the full KLayout sky130 deck on 6_final.gds
G-09	Physical verification	Independent LVS	Reference CDL comes from the same 6_final.odt	Magic extraction + Netgen LVS against a netlist from 6_final.v
G-10	Equivalence	Formal RTL to sky130 netlist equivalence	Generic netlist only; kepler-formal compares netlist stages	eqy or synth/equiv_cut.py with gold = RTL, gate = sep_1_2_yosys.v/6_final.v and sky130 cell functions
G-11	Power	Activity-based power, power at worst corner	Default activity only (K-06)	VCD/SAIF from a representative workload (directed bench or GLS); read_vcd/read_saif in OpenSTA; SS/FF power
G-12	Power integrity	EM (grid and signal)	Not run; scratch currents only, no limits	sky130 EM current-density limits per layer/via; PSM -enable_em with activity-based current; signal-net EM
G-13	Power integrity	Dynamic IR, decap budget, package/bump model	Static IR only; no decap cells	Vectored PSM with switching activity; decap insertion and model; package/bump RLC
G-14	Package / IO	Pad ring, IO cells, ESD, supply pads, seal ring, package, bonding diagram, pin-to-lead map	Core-only block (K-05); brief: package undecided	Choose package and sky130 IO library; decide pad count vs 215 signals (IO multiplexing or pad-limited die); pad frame; re-run DRC/LVS/STA with IO cells
G-15	Package / IO	Absolute maximum ratings, recommended operating conditions, IO DC/AC levels, ESD/latch-up ratings	No such data in any file	IO cell datasheet for the chosen library; supply and temperature ranges
G-16	Package / IO	Intended pin assignment	Pins are place_pins output; buses split across edges	Pin constraints (set_io_pin_constraint/place_pin) and a re-run
G-17	Power intent	Statement of single-domain intent	One VDD/VSS domain; brief proposes power-domain isolation for the concept	SPEC line "single domain" (N/A for UPF) or UPF if domains are planned
G-18	Fault coverage	Injection on comparators, mismatch OR-tree, voter, nxt, in_fire, clr, clock/reset leaves, sensor/handshake inputs; MBU; same bit in both copies; product in the campaign	Not injected (K-03)	Transient/stuck-at scenarios in fault/gen_fault_copies.py and force/release in tb_seu_campaign.v with expected outcomes
G-19	Fault coverage	fault_q 1→0, stickiness and fault -> !out_valid && !in_ready invariant; out_data from copy A every cycle; code-3 behaviour	Single-upset model cannot reach it; m07, m20, m22, m27 survive (K-10)	Double-fault scenarios; the checks of tb_mutation_gaps.v T1-T4 moved into the fault harness or sim suite
G-20	Fault coverage	Gate-level/post-layout fault injection incl. the 3 unprotected flops, per-flip records and a golden scoreboard	GLS injects 518 redundant bits only, aggregates only, lockstep without golden reference	Extend pd/gls/gl_inject.vh and tb_gls.v: unprotected flops, per-injection \$display, uninjected golden instance
G-21	Radiation	Physical basis for separation and any upset rate (cross-section, FIT)	Geometry only; sky130 not characterized (K-08)	Beam test or TCAD/charge-sharing data for sky130; outside the open flow
G-22	Statistics	Multi-seed campaign and GLS, confidence intervals	Campaign seed 1, GLS seed 1 / 30000 cycles	+seed=N runs; per-group rates with intervals
G-23	Simulation	Line/branch/toggle coverage	Testbench event counting only	verilator --coverage (or equivalent) on directed and cocotb runs, merged report

ID	Area	Missing item	Why (evidence)	Needed to produce it
G-24	Simulation	cocotb random test on a netlist; mid-run reset coverage points	Only the directed bench runs on netlists; seed 2 has no mid-run reset	<code>run_random.py --sources <sep netlist> <cells.v></code> ; <code>reset-in-THROTTLE/STOP/out_fire</code> as REQUIRED points
G-25	Simulation	Netlist-level waveforms	Package waves are RTL-only (hierarchical RTL paths)	Map injections to netlist instance names (as <code>pd/gls/gen_gls.py</code> does)
G-26	Provenance	Pinned toolchain and PDK per run	OpenROAD "unknown", ORFS commit and PDK version unrecorded, floating tag (K-09)	Pin the image by digest in <code>mk/pd*.mk</code> ; log <code>openroad -version</code> , ORFS commit, <code>open_pdk</code> s version, liberty hashes, simulator and kepler-formal versions
G-27	Provenance	RTL/TB hashes and netlist identity in formal, sim and GLS outputs	Formal says "RTL: rtl"; sim logs no hash; GLS logs identical across netlists (D-24, D-44)	<code>sha256</code> of inputs and <code>git rev-parse HEAD</code> in each summary
G-28	Provenance	Logs for hand-run and sabotage claims	Formal/fault "tried once by hand", ECC solver attempts, <code>review_sabotage.txt</code> have no logs	Re-run with logs or mark as CLAIM (unverified)
G-29	Published evidence	Sep directed-bench GLS, sep GLS mutant, re-published storage audit (19/19), report console log, in reports/	Exist only as package re-runs (K-11, K-15, D-55)	<code>make sim-gls SIM_GLS_SRCS=<sep 6_final.v></code> ; add <code>pd-sep-gls-mutant</code> ; re-run <code>pd-sep-report</code> with logging
G-30	Published evidence	Full formal logs, traces, per-trial campaign data, generic netlist and full synthesis log	Trimmed logs and aggregates only; build/ is gitignored	Copy <code>build/formal/**/trace*.vcd</code> , <code>build/fault/**/trace.vcd</code> , <code>build/fault/campaign/trials.tsv</code> , <code>build/ecc/formal/**/*.*.log</code> , <code>build/synth/orbit_demo_synth.v</code> , <code>build/synth/synth.log</code>
G-31	Documentation	Top-level README with revision, prerequisites and make targets	Referenced but absent (D-10)	Owner-written README
G-32	Documentation	Revised brief and SPEC with change log	Both v0.1 at 26566f0 (D-07 to D-09)	Brief/SPEC revision citing 07_verification results
G-33	Documentation	Demonstrator throughput at the closed clock	Model never reads a closed clock (D-56); peak at 138.9 MHz = 1.1111 GOPS is arithmetic only	Fix <code>MODEL_PD_SUMMARY</code> ; a throughput testbench for measured rate
G-34	Design decisions	Disposition of research-note MUST items (R1-R3, P1, P3, V1-V4)	None implemented; SPEC has no change log	Owner decision per item: accept in SPEC 8 or schedule an RTL/PD change
G-35	ECC	Integration, requirements, PDK-mapped timing/area/power, fault model beyond the array, other parameter sets, machine-checked end-to-end proof, negative control for wrong check bits	Standalone block (K-12); unbounded proofs abstract the decoder; DEPTH 16/INTERLEAVE 2 only	Integration spec and RTL; SPEC requirements (scrub interval, error rates); ORFS run with SDC; fault injection on control registers; deeper real-decoder BMC or k-induction; a <code>wr_cw[71:64]</code> mutant
G-36	Reference runs	IHP SG13G2 sign-off	DRC FAIL (5 density markers), LVS and LEC not run (reference only, N/A for the reviewed layout)	Only if the IHP data point is to be reviewed: fill tuning, LVS setup, non-vacuous LEC
G-37	Presentation	Sources for the capability matrix; story-page render inputs; check that the published explorer artifact matches <code>viz/index.html</code>	Unsourced cells (D-70); renders from an external directory; artifact link not compared	Cited evidence per cell or removal; commit the render command; compare artifact with <code>viz/</code> and the <code>layout.json</code> hash
G-38	Silicon	Fabricated parts and bench measurement	Not fabricated (brief p.1)	Pad ring, sign-off tapeout, fabrication, lab characterization